

ZamiaCAD, an Open Source Platform for Advanced Hardware Design

ZamiaCAD is a modular and extensible platform for your advanced hardware design, analysis and research. Let us take a look at what it delivers

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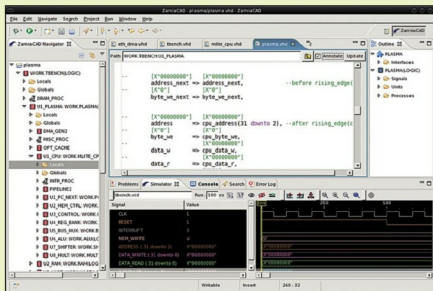
If you are looking for an open source application for real-world hardware design, debug and analysis, ZamiaCAD could be the right solution for you. ZamiaCAD has been implemented with high performance, scalability and usability as the main targets, and it has already become a basis for VHDL IDE Eclipse plug-in with built-in simulator.

Significant advances have been made in software development discipline in the area of program code entry, navigation and analysis, while the existing hardware development environments are still lacking this progress, especially in the open source tools domain. ZamiaCAD is an open source platform that aims at transferring the advances in software development to the hardware development discipline.

How it works

This multi-platform tool provides a modular and extensible framework that can also be used for design entry, navigation and analysis along with the simulation functions. Besides, the cache and persistent data structures make this platform capable of handling very large industrial designs.

If we look at the structure of ZamiaCAD, it can be broken into various building blocks, such as frontend, its core and an Eclipse IDE plug-in based GUI, with various features providing a handy environment for the



ZamiaCAD

advanced hardware designing.

The language-dependent modular and extensible frontend for HDL parsing consists of a parser and an elaboration engine, which help design engineers utilise the advances in software development in the hardware development discipline.

Currently, full VHDL elaboration is included with a complete frontend.

But Verilog has only a parser with persistent and scalable syntax tree storage backing both these frontends, which helps these language-dependent frontends to generate language-independent instantiation graphs (IGs), accounting for the scalability of intermediate design representations in terms of size and performance.

Responsible for the intermediate

An interpreter for elaboration

Computation of various static values, like the range of boundaries for the system type or conditions and the loop ranges at the time of unrolling generate statements, is done at the time of elaboration. For easy and fast computations, ZamiaCAD includes a complete interpreter implemented as a stack machine supporting all operations that can occur during elaboration, which include math and logic operators and even (potentially recursive) subprogram calls. This quick expression-evaluation interpreter is implemented in the package *com.zamiacad.instgraph.interpreter*.